

TASK 2: SQL Complex Queries Solution

This document provides SQL scripts demonstrating join operations, subqueries, and aggregate functions for extracting business metrics from a sample e-commerce database.

Sample Database Schema

- customers (customer_id, name) - orders (order_id, customer_id, order_date) - order_items (order_id, product_id, quantity, unit_price) - products (product_id, product_name)

SQL Queries

1. INNER JOIN: Total Revenue per Customer

```
SELECT c.customer_id, c.name, SUM(oi.quantity * oi.unit_price) AS total_revenue FROM customers c
INNER JOIN orders o ON c.customer_id = o.customer_id INNER JOIN order_items oi ON o.order_id =
oi.order_id GROUP BY c.customer_id, c.name ORDER BY total_revenue DESC;
```

2. LEFT JOIN: Customers with No Orders

```
SELECT c.customer_id, c.name, COUNT(o.order_id) AS order_count FROM customers c LEFT JOIN
orders o ON c.customer_id = o.customer_id GROUP BY c.customer_id, c.name HAVING
COUNT(o.order_id) = 0;
```

Outcome: Professional SQL skills for complex data analysis and business metrics extraction.